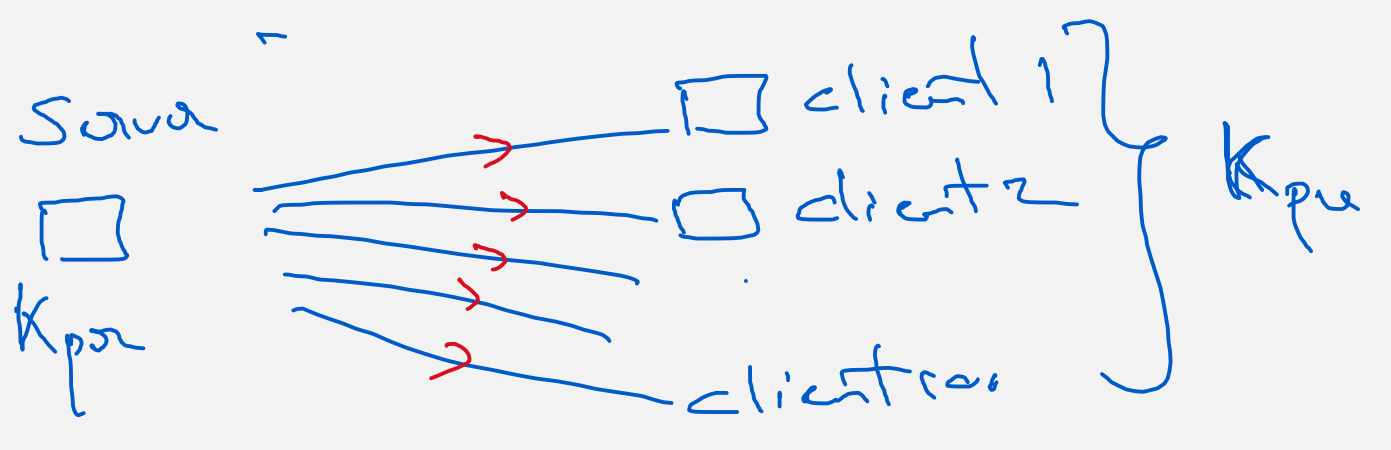
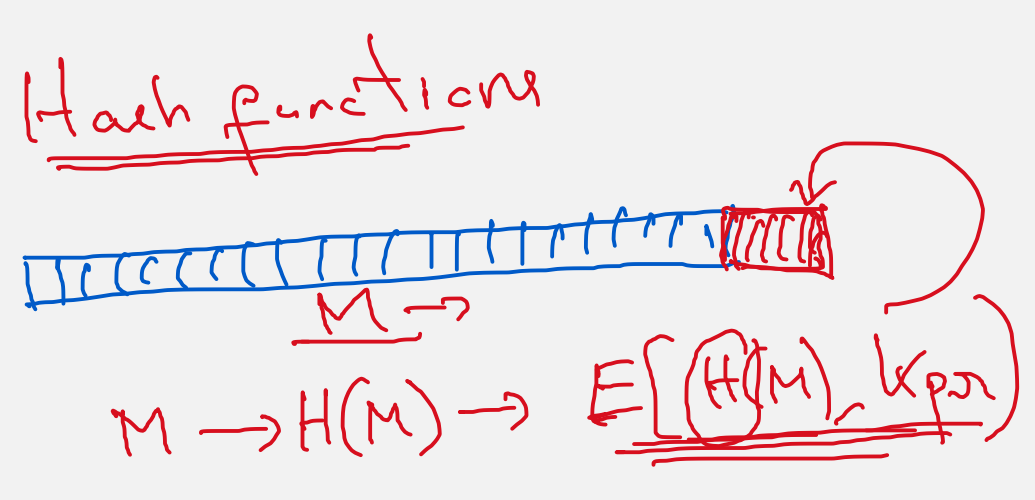
Authentication mechanism

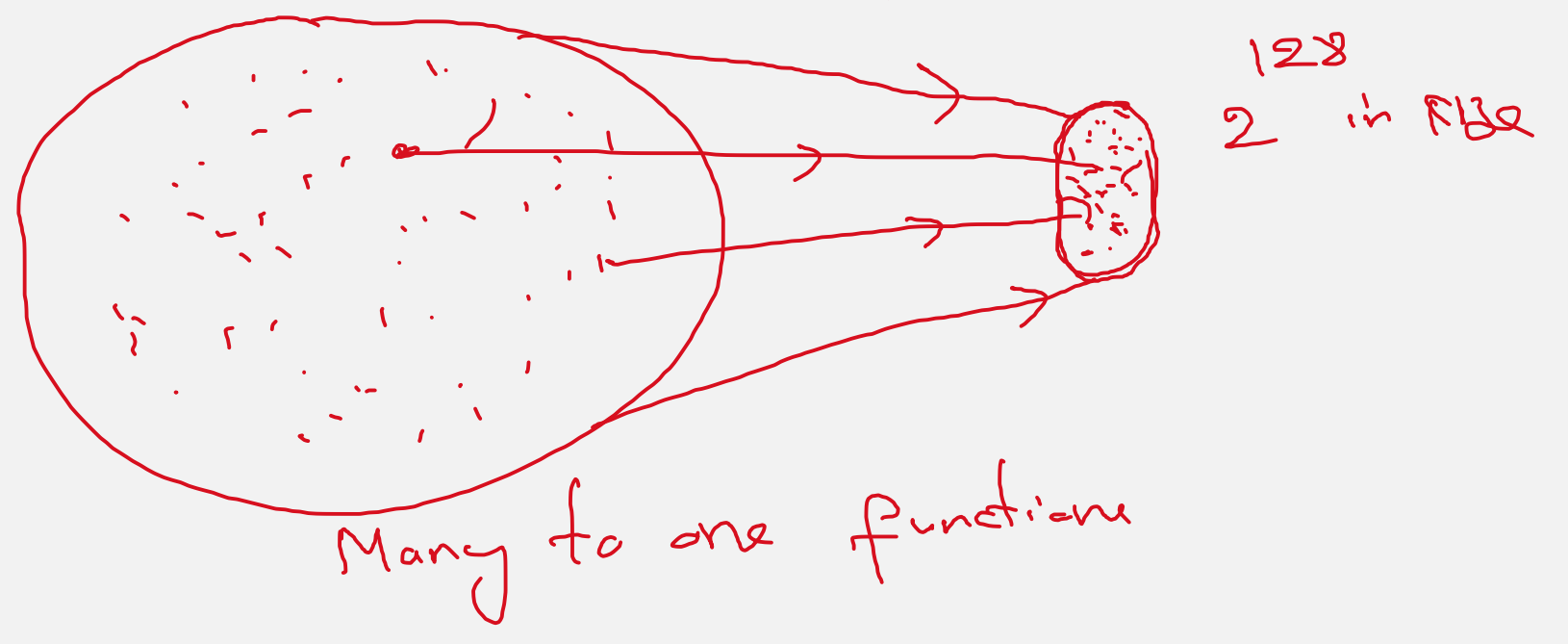

Server
Kpr
client 1
client 2
client 100
Kpu

1) Hash functions



$M \rightarrow H(M) \rightarrow E[H(M), K_{pr}]$

Hash functions


128 bit
2^{128} in size
Many to one functions

Example

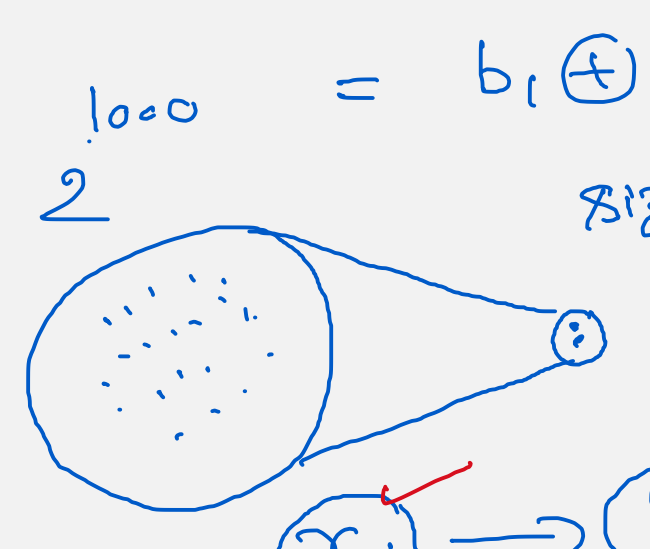
Input is a binary vector of length 1000

$M = [b_1\, b_2\, b_3 \ldots b_{1000}]$
first bit, last bit

$H(M) = \left(\sum_{i=1}^{1000} b_i\right) \bmod 2$

$= b_1 \oplus b_2 \oplus b_3 \ldots + b_{1000}$

size of out is 1 bit



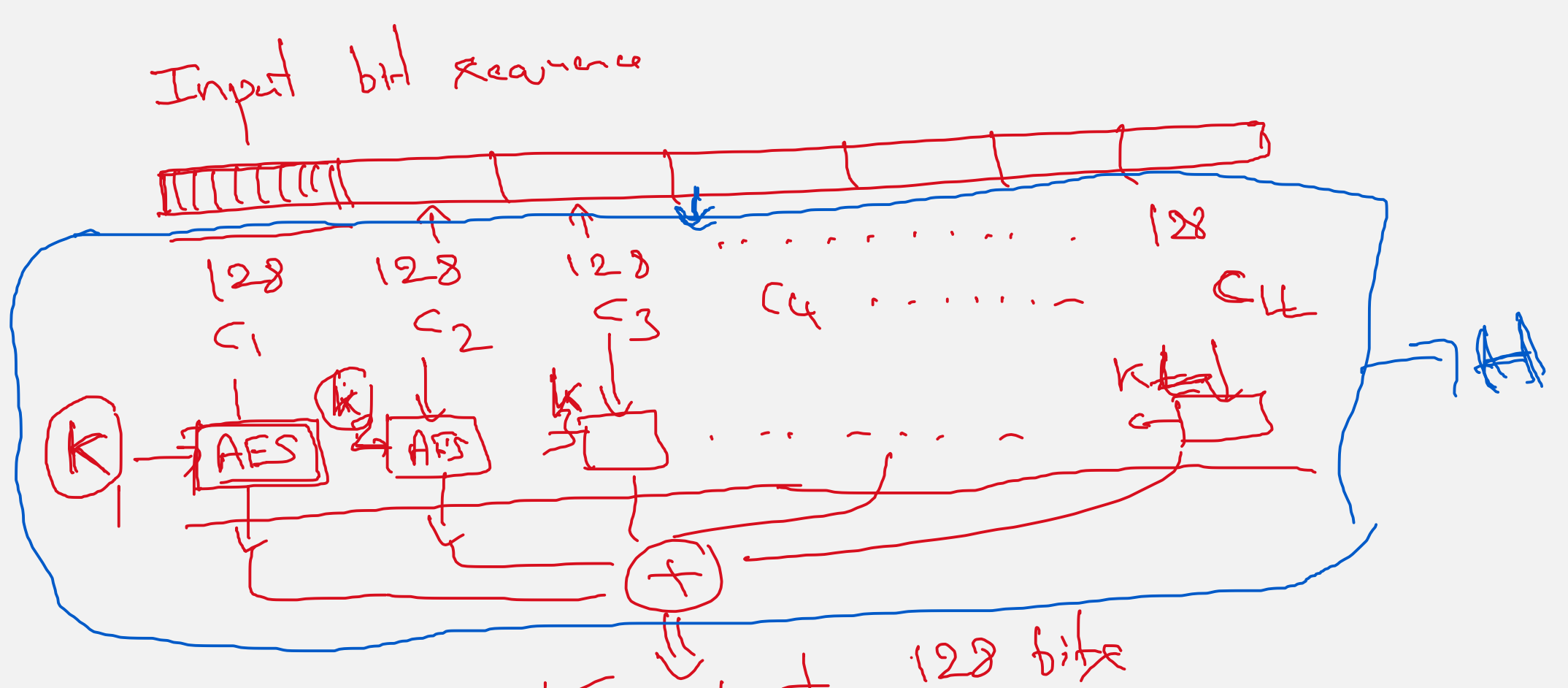
$x_1 \rightarrow H(x_1)$

$x_2 \neq x_1$ s.t. $H(x_2) = H(x_1)$

x_1 [1111111 ... 11] → 0

x_2 0011111111 ..... 11] → 0

Input bit sequence


128
128
128
128
c1
c2
c3
c4
cL
K
K
K
K
AES
AES
+
H
output
128 bits

$x_1 \rightarrow H(x_1)$

x_2 s.t. $H(x_2) = H(x_1)$

1) MD5 → M → 128 bit
2) SHA-1 M → 160 bit
3) SHA-2 M → SHA256
224, 384, 512

NIST → National institute of standards and technology

Question

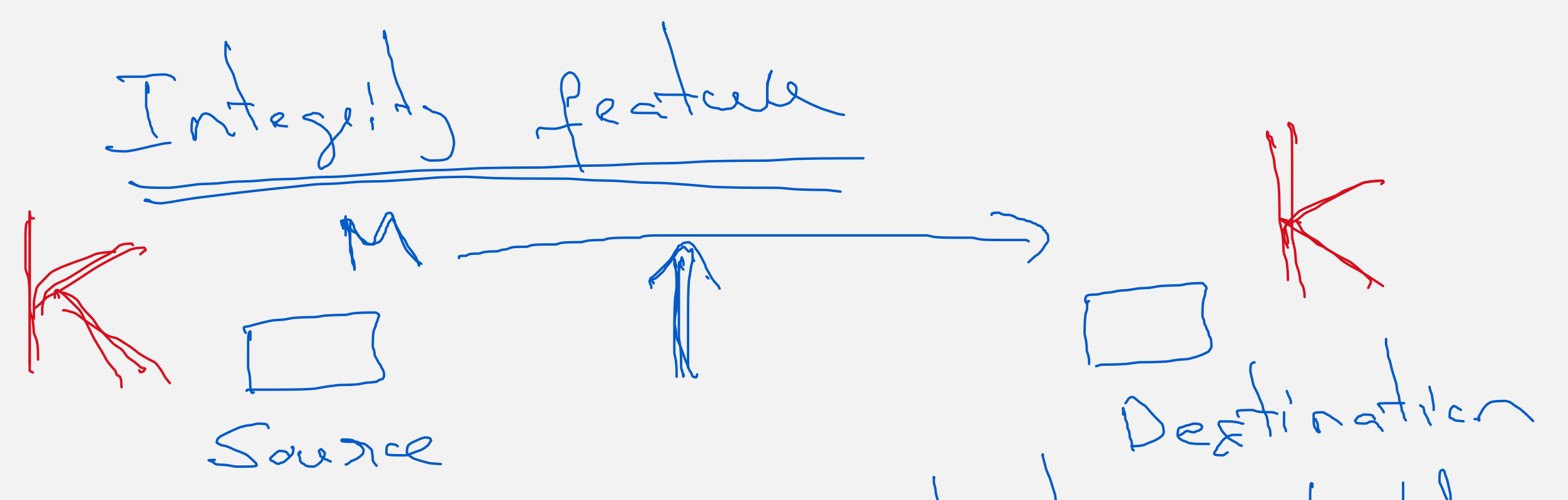
Ca I use a good Hash function as a symmetric key cipher?

private ← 1) Confidentiality → DES, AES (symmetric) / RSA (public key crypto)

who? ← 2) Authentication → RSA + Hash function (SHA-2, SHA-3)

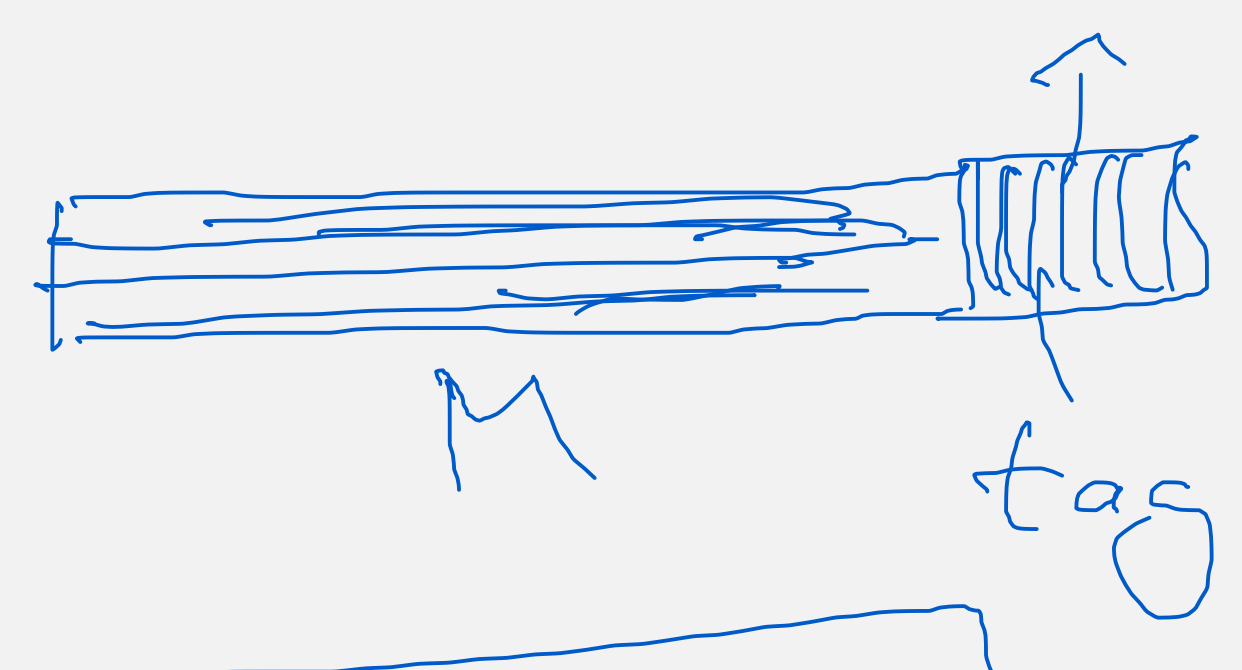
integrity? 3) Message authentication code (MAC)

Integrity feature


K
M
Source
K
Destination

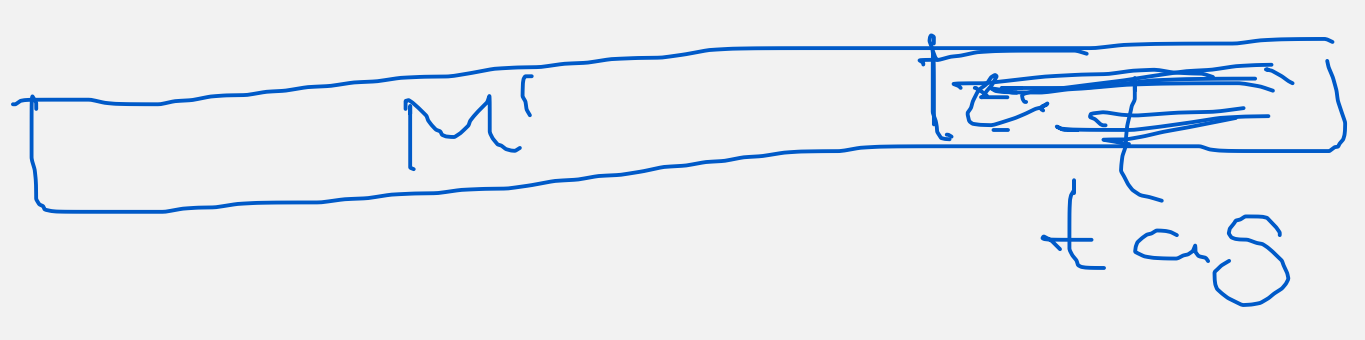
Destination should know if a third party has modified the data

CRC

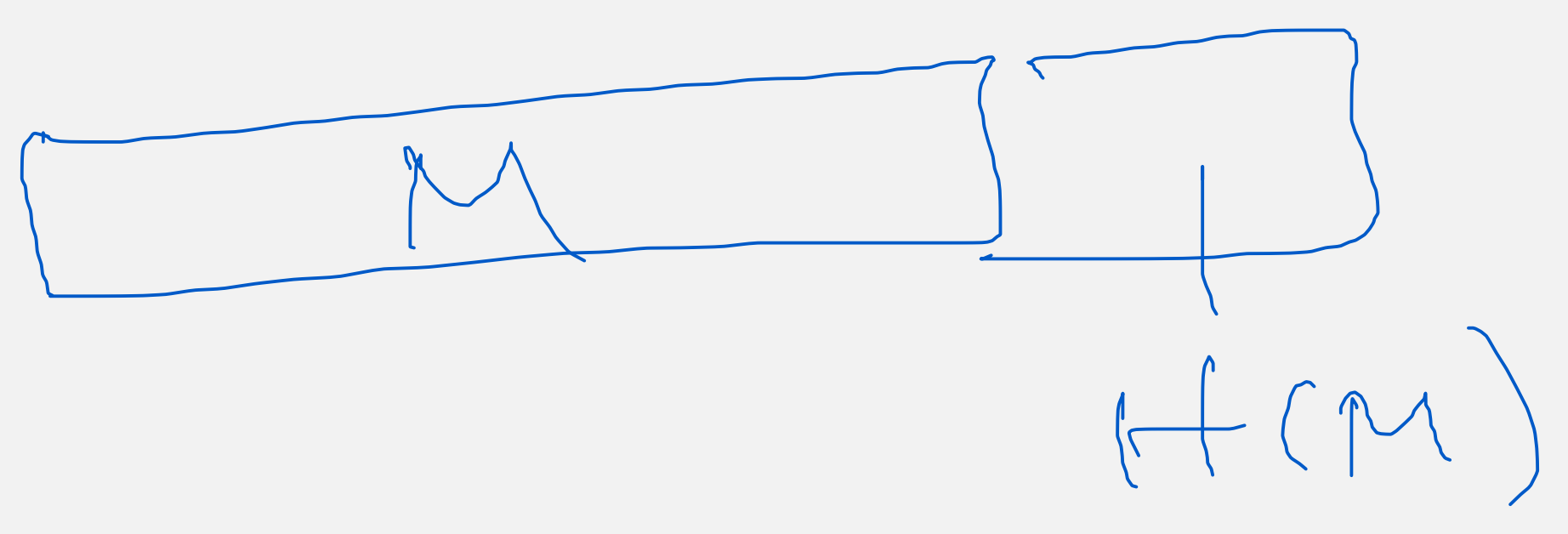


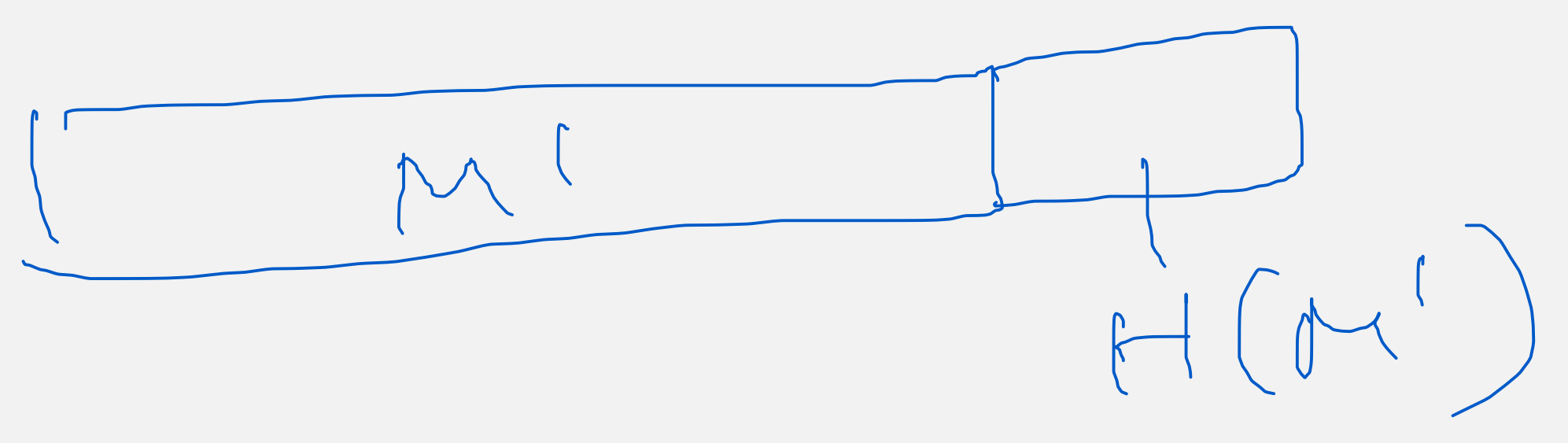
tag = f(M)
linear combination

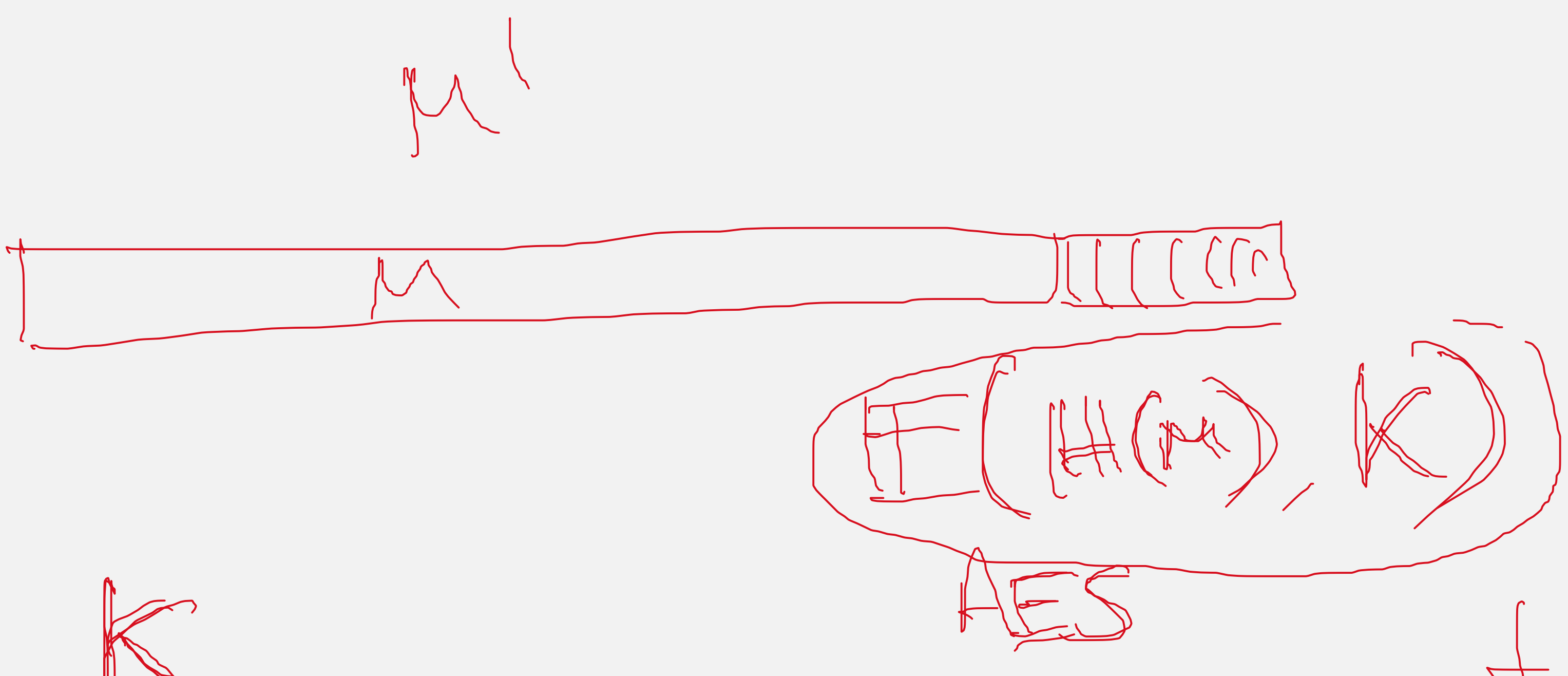
$\boxed{M \Rightarrow M'}$

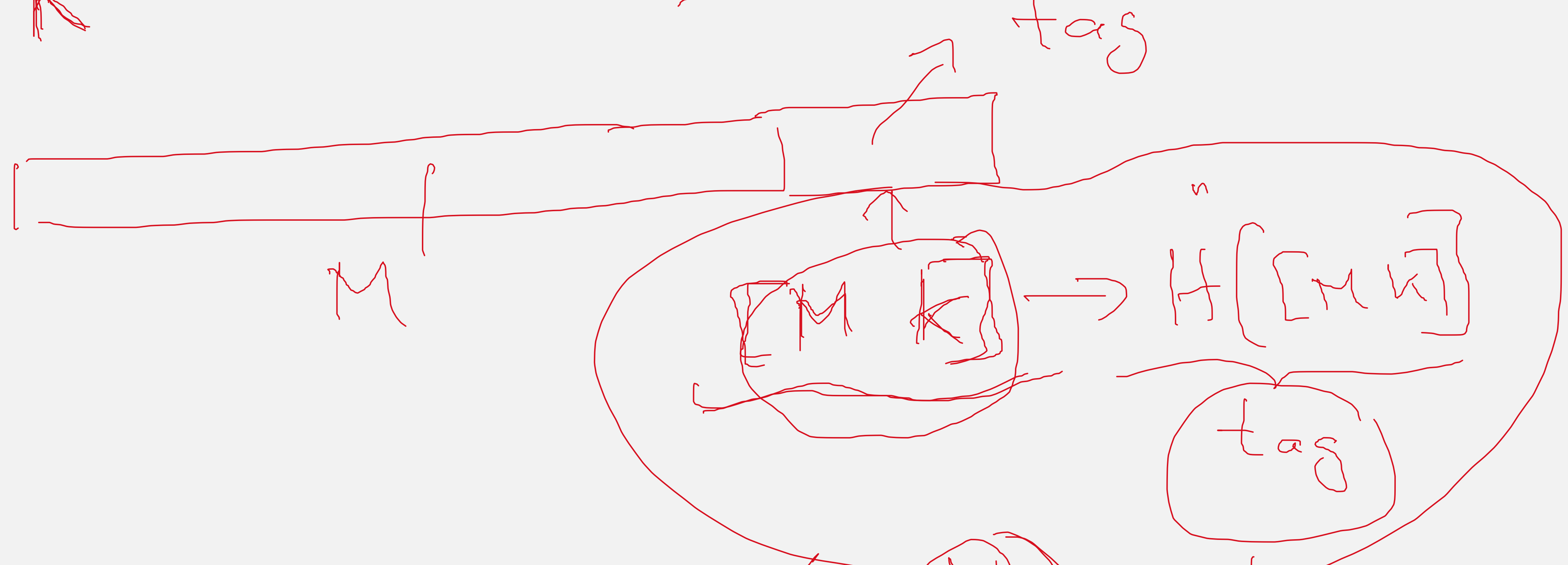


$f(M') \neq$ tag






M'
M
E(H(M), K)
AES
K


tag
M
[M K] → H[M K]
tag

AES(M, K) → tag

AES(H(M), K)